Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/02/2010
PI Name:	Azzam, Edouard Ph.D.		
Project Title:	The Role of Gap-Junction Communication and Oxidative Metabolism in the Biological Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Newark	**State:**	NJ
Zip Code:	07103	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date extended until 12/31/2011, per C. Guidry/JSC (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Toledo , Sonia (UMDNJ-New Jersey Medical School) Harris, Andrew (New Jersey Medical School (UMDNJ)) Howell, Roger (University of Medicine and Dentistry of New Jersey) Pain, Debkumar (UMDNJ - New Jersey Medical School)		
Grant/Contract No.:	NNJ06HD91G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The objective of this project is to investigate biochemical and molecular aspects of two inter-related fundamental biological processes, namely gap-junction intercellular communication and oxidative metabolism, in determining human cellular responses to exposures of very low to moderate fluences of protons (of energies such as found in the space environment) or high charge (Z) and high energy (E) particles (1 GeV/nucleon 56Fe ions). Gap junctions are found between most cells in the body, and the molecular signaling they mediate is crucial for normal development, physiology and response to disease. By allowing direct intercellular transfer of cytoplasmic molecules, they provide a powerful pathway for direct molecular signaling between cells. On the other hand, oxidative metabolism is primarily regulated by mitochondria, which mediate respiration and ATP synthesis in eukaryotic cells. They also participate in numerous indispensable metabolic pathways. We hypothesize that exposure of human cells to high or low fluences of protons and/or HZE particles results in distinct molecular and biochemical changes not only in irradiated but also neighboring bystander cells. These events are modulated by oxidative metabolism and gap-junction coupling, which together induce persistent changes in mitochondria and in nuclear DNA in progeny of irradiated and affected bystander cells. In Aim 1, we propose to define the aspect(s) of junctional communication leading to proton- and high charge/high energy (HZE) particle-induced toxic effects. Specifically, we will examine the role of different connexin proteins in propagating death-inducing (growth delay, apoptosis) or pro-survival effects (proliferative capacity) among irradiated cells and also between irradiated and non-irradiated cells. Aim 2 is to identify specific connexin channel permeabilities that correlate with intercellular propagation of toxic or protective effects in cultures exposed to HZE particles or protons. An emphasis will be to determine the nature of molecules communicated among irradiated cells and between irradiated and non-irradiated cells. In Aim 3, we will evaluate mitochondrial function in cells exposed to low fluences of high energy charged particles. Specifically, we will examine mitochondrial protein import, mitochondrial protein synthesis, assembly of large protein complexes including respiratory complexes, and oxidative damage to mitochondrial proteins and DNA. We will also evaluate the persistence of oxidative effects in the progeny of exposed cells and correlate the observed results with the overall cellular redox environment.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The proposed experiments relate to Risk No. 28, subsection 28b (molecular studies on carcinogenesis) in NASA's 'Roadmap'. In addition to revealing signaling pathways underlying non-targeted effects of ionizing radiation, the outcome of this proposal may provide insight into space radiation-induced degenerative diseases such as cognitive defects and cardiovascular disorders (Risks No. 29 and 30). It will provide a mechanistic basis for understanding exposures and risks to human from low fluences of high and low linear energy transfer radiations. By identifying the nature of molecules communicated among irradiated and neighboring non-irradiated cells, it may contribute to remediation strategies against harmful effects of space radiation, and is also relevant to heavy ion and proton radiotherapy.		

Task Progress:	During this reporting period, we have extended our in vitro research on the role of intercellular communication and oxidative metabolism in the biological responses to energetic protons and iron ions to other particulates radiations, namely silicon ions, titanium ions and carbon ions. In collaboration with Dr. Ruth Globus and Bernard Rabin, we have also extended our characterization of the mechanisms underlying the targeted and non-targeted biological effects of space radiation from in vitro to in vivo systems. We have shown that similar to the observations made in vitro by our laboratory and others, non-targeted effects of space radiation do exist in vivo. Stressful effects in non-irradiated tissues were detected months after cranial irradiation with high charge and high energy particles. The induced effects involve a prominent role for oxidative metabolism and inflammatory responses. Using several types of particulate radiations, our in vitro studies have confirmed that the linear energy transfer (LET) of the radiation is critical in the propagation of stressful effects from irradiated to neighboring non-irradiated bystander cells. Further, high LET-induced stressful effects in bystander cells are transmitted to their progeny and can lead to an increase in the frequency of spontaneous transformation from the normal to the neoplastic state. The nature of the molecules mediating the propagation of stressful biological effects from irradiated to bystander cells remains under investigation; a particular focus of our ongoing research is on the role of gap junctions in the transmission of the mediating molecules from irradiated to bystander cells. With further relevance to risk assessment, we have shown in experiments using mixed beams of space radiation that bystander cells co-cultured, in vitro, with cells exposed to low doses from energetic protons are protected from the damaging effects of a subsequent challenge dose from energetic iron ions. The protective effect is transient and lasts for about 24 h. Several repeat experiments to confirm observations made in earlier reporting periods were achieved, which now allows us to prepare a large amount of the generated results for publication.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
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